ADITYA SHIVDATTA JARE

AI/ML Engineer · Japanese Language (N3, in progress)
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PROFESSIONAL SUMMARY

AI/ML Engineer with a strong foundation in Machine Learning and Deep Learning, including hands-on work with CNNs and working knowledge of RNNs. Experienced in designing and evaluating Retrieval-Augmented Generation (RAG) pipelines, with a fine-tuned domain-specific classification model that improved accuracy by 6.44%. Comfortable across the core ML workflow — data preparation, model training and fine-tuning, evaluation, and deployment — with solid Python fundamentals and working exposure to cloud tools (Azure). Certified in Japanese (NAT-N4) and currently studying for N3, bringing the ability to support QA and evaluation of Japanese-language model outputs for grammatical and cultural accuracy.

TECHNICAL SKILLS

- Machine Learning & Deep Learning: Supervised learning, model evaluation and metrics, class-imbalance handling, ANN, CNN, transfer learning, RNN, Keras functional API, hyperparameter tuning (Keras Tuner, GridSearchCV), TensorFlow, PyTorch, scikit-learn, XGBoost
- NLP & RAG Systems: RAG architecture, transformer fine-tuning (Legal-BERT), vector databases (Pinecone, ChromaDB), prompt engineering
- Programming & Tools: Python (NumPy, Pandas, Matplotlib), SQL, Git/GitHub, Linux/Bash basics, Docker, Streamlit, FastAPI
- Cloud Fundamentals: Microsoft Azure (Azure Translator, Blob Storage, Cosmos DB, Linux VMs, RBAC security)

PROJECTS

JURI-AI: AI-Powered Legal Assistant Platform (RAG System) | *GitHub* | *Live Demo*
- Built a RAG pipeline for legal queries (89% Q&A accuracy) with sub-100ms semantic search on 768-dim embeddings
- Fine-tuned Legal-BERT on 10,000 contract clauses (15 classes), improving accuracy by 6.44% to 84.76%
- Wrote an evaluation framework to document failure modes, and presented the architecture in formal Japanese

FinPulse: AI Financial & Earnings Call Research Copilot | *GitHub* | *Live Demo*
- Built a RAG copilot for SEC 10-K/10-Q filings and earnings calls that answers with source citations
- ChromaDB search plus BGE reranking cuts hallucinated claims; Groq flags guidance inconsistencies with confidence badges
- Deployed on Streamlit Community Cloud with sub-2 s query latency

Age and Gender Estimation from Faces (CNN, Transfer Learning) | *GitHub*
- Two-head VGG16 transfer-learning model (Keras functional API) that predicts age and gender from UTKFace images
- On 2,371 held-out images: age MAE 6.5 years (79.5% within 10 years), gender accuracy 93.3%, ROC-AUC 0.98

RESEARCH PUBLICATION

- JURI AI: AI System for Legal Assistance and Lawyer Appointment Management — International Journal of Innovative Research in Technology (IJIRT), Volume 13, Issue 1 (June 2026) | ISSN: 2349-6002
- Authors: S. Tikole, A. Jare, R. Margale, F. Mistry | *Paper link*

EDUCATION

Bachelor of Engineering (B.E.) in Artificial Intelligence & Machine Learning | *P.E.S. Modern College of Engineering, Pune | Graduation: 2026*
- CGPA: 8.77 / 10
- Relevant coursework: Machine Learning, Deep Learning, RAG, Cloud Computing

CERTIFICATIONS

- Machine Learning Specialization, Stanford University & DeepLearning.AI (Coursera). Credential ID: 7FMBS1YVA6QD | Verify
- Neural Networks and Deep Learning, DeepLearning.AI (Coursera). Credential ID: 326c383fbebc9c078b62895950e486ae | Verify
- Improving Deep Neural Networks, DeepLearning.AI (Coursera) | Verify
- Convolutional Neural Networks, DeepLearning.AI (Coursera). Credential ID: 18eeee6c171a13d5bd69826180cba514 | Verify

JAPANESE LANGUAGE PROFICIENCY

- NAT-TEST N4 certified; preparing for JLPT N3 (intermediate kanji, grammar patterns, and technical AI vocabulary)
- Co-hosted a Japanese cultural exhibition booth at Bunkasai

LANGUAGES & INTERESTS

- Languages: Japanese (NAT-N4, studying N3), English (fluent/professional), Hindi (native), Marathi (native)
- Interests: computer vision, Japanese linguistics and culture, horse riding, music listening